

Prove polynomial-time convergence of Scarf's algorithm for any ordinal matrix setting

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Website #10 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

I interpret the input polytope as the standard unit-capacity bipartite matching polytope with loops:

$$P(G) = \{x \in \mathbb{R}_{\geq 0}^{n+m} : Ax = \mathbf{1}\},$$

where $G = (U \sqcup W, E)$ is bipartite, $n = |U| + |W|$, the first n columns of A are the loop columns I_n , and the remaining $m = |E|$ columns are the node-edge incidence vectors of the valid edges. Let C be an arbitrary ordinal matrix.

The question is whether there is a polynomial p such that for every such input $(A, \mathbf{1}, C)$, Scarf's algorithm can be made to terminate after at most $p(n+m)$ iterations (one iteration = one cardinal pivot, followed unless already terminated by one ordinal pivot), perhaps by a suitable rule for resolving degeneracy in the cardinal pivot.

I do *not* settle the full question. What I prove is the following strongest rigorous partial result I found:

If the bipartite graph G has bounded matching number $\nu(G)$, then for every ordinal matrix C there exists a terminating execution of Scarf's algorithm whose number of iterations is polynomial in $n+m$. In fact,

$$T \leq n(n+m) \sum_{k=0}^{2\nu(G)} \binom{m}{k} 2^k \leq (2\nu(G)+1)2^{2\nu(G)}(n+m)^{2\nu(G)+2}.$$

Thus the existential version of the problem is polynomial for every fixed value of $\nu(G)$; in particular, when one side of the bipartition has constant size.

Alternative readings considered

- **Reading A (stronger).** There exists a *single*, explicit, instance-independent cardinal pivot rule under which every input $(A, \mathbf{1}, C)$ on a bipartite matching polytope terminates in polynomially many iterations.
- **Reading B (weaker, and the one explicitly formalized in the prompt).** For every input, there exists *some execution* (equivalently, some way of resolving the cardinal-pivot choices along that run) that terminates in polynomially many iterations.
- **Capacity variant.** One might also read the problem as allowing arbitrary positive right-hand sides b , i.e. a bipartite b -matching/transportation setting rather than the unit-capacity matching polytope.

The literature I found treats the exact arbitrary- C question as open; I therefore retreat from Reading A to Reading B, and I keep the standard stable-marriage reading $b = \mathbf{1}$. My proof does not cover arbitrary capacities.

Related literature found

- **Faenza–He–Sethuraman (2025), *Scarf’s Algorithm and Stable Marriages*** — PARTIAL + TOOL. This paper proves polynomial-time convergence for a *specific* consistent matrix C^* attached to stable-marriage preferences, proves structural lemmas on bases of the bipartite matching polytope (their Lemmas 21–22), and explicitly identifies the arbitrary-ordinal-matrix question on the bipartite matching polytope as an interesting open problem. (pubsonline.informs.org)
- **Chandrasekaran–Faenza–He–Sethuraman (ICALP 2025), *Scarf’s Algorithm on Arborescence Hypergraphs*** — PARTIAL. This gives polynomial-time convergence on arborescence hypergraphs for a specially designed block-partitioned ordinal matrix, and explicitly says they are not yet able to prove polynomial convergence for the larger class of network hypergraphs. This is a promising near-neighbor but does not address arbitrary C on bipartite matching polytopes. (drops.dagstuhl.de)
- **Kintali–Poplawski–Rajaraman–Sundaram–Teng (2013), *Reducibility among Fractional Stability Problems*** — BACKGROUND. This proves the general SCARF problem PPAD-complete, so unrestricted Scarf instances are computationally hard; however, it does not specialize to the bipartite matching polytope with arbitrary C . (epubs.siam.org)
- **Scarf–Hansen (1973), *The Computation of Economic Equilibria*** — BACKGROUND. Classical source for Scarf’s pivoting method and its perturbation-based handling of degeneracy; I use this only as background for the existence of at least one terminating execution on degenerate inputs. The more recent Faenza–He–Sethuraman paper explicitly summarizes this perturbation viewpoint in §3.3. (cowles.yale.edu)

A promising route that *did not* immediately apply was to try to import exponential lower bounds from network simplex. The ICALP 2025 paper remarks that a related leaving rule for network simplex can still suffer exponentially many degenerate pivots when entering variables are unrestricted; however, this does not directly translate to Scarf’s setting because Scarf’s entering column is constrained by the ordinal pivot and the paper itself does not obtain a lower bound for Scarf on network hypergraphs. (drops.dagstuhl.de)

Proof sketch

The proof is a counting argument parameterized by the graph matching number

$$r := \nu(G).$$

The key combinatorial fact is that although a feasible basis B of the bipartite matching polytope may be highly degenerate, it cannot contain too many non-loop columns. Indeed, classical structure theory says that the valid-edge part of B is a forest and every connected component carries exactly one loop. The corresponding basic feasible solution is integral; hence its positive valid-edge columns form an honest matching M_B . In a tree component with one loop, if s valid edges of the basic solution are equal to 1, then the component has $2s + \lambda$ vertices, where $\lambda \in \{0, 1\}$ indicates whether the loop itself is selected by the basic solution. Since a tree on $2s + \lambda$ vertices has $2s + \lambda - 1 \leq 2s$ valid edges, each component contributes at most twice its number of matched edges. Summing gives

$$|B \setminus [n]| \leq 2|M_B| \leq 2\nu(G).$$

So every feasible basis uses at most $2r$ non-loop columns. Once the valid edges are fixed, the only remaining freedom is: in each nontrivial tree component, which vertex carries the unique loop of that component. A component with s valid edges has $s + 1$ vertices, hence at most $s + 1 \leq 2^s$ such choices. Therefore the number of feasible bases with exactly k non-loop columns is at most $\binom{m}{k} 2^k$, and summing over $k \leq 2r$ gives a polynomial bound for fixed r .

A Scarf pair (B, D) with $B \neq D$ is determined by a feasible basis B , a leaving column $j_\ell \in B$, and an entering column $j_t \notin B$, since then $D = B - \{j_\ell\} \cup \{j_t\}$. Hence the number of possible Scarf pairs is at most

$$n(n+m) \sum_{k=0}^{2r} \binom{m}{k} 2^k.$$

Finally, classical perturbation theory for Scarf's algorithm guarantees that at least one finite terminating execution exists even in the degenerate case. Among all terminating executions, choose one with the minimum number of iterations. That execution cannot repeat a Scarf pair: if $(B_i, D_i) = (B_j, D_j)$ with $i < j$, then the segment between i and j can be excised, producing a shorter terminating execution. Therefore the length of a shortest terminating execution is at most the number of possible Scarf pairs, yielding the announced bound.

Main result

Theorem 1. *Let $G = (U \sqcup W, E)$ be a bipartite graph, let $A = (I_n \mid A_G)$ be the standard-form matrix of its unit-capacity matching polytope with loops, let $b = \mathbf{1}$, and let C be an arbitrary ordinal matrix. Write*

$$N := n + m, \quad r := \nu(G).$$

Then there exists a terminating execution of Scarf's algorithm on $(A, \mathbf{1}, C)$ with at most

$$nN \sum_{k=0}^{2r} \binom{m}{k} 2^k \leq (2r+1) 2^{2r} N^{2r+2}$$

Scarf iterations.

Consequently, for every fixed r , the existential version of the problem is polynomial on the class of bipartite graphs with $\nu(G) \leq r$. In particular, if one side of the bipartition has size at most r , then such a polynomial bound exists.

Proof

Lemma 1. *Let B be a feasible basis of the bipartite matching polytope $P(G)$, and let x^B be its basic feasible solution. Then the set*

$$E_B := B \cap E$$

of non-loop columns satisfies

$$|E_B| \leq 2\nu(G).$$

Proof. I invoke two structural lemmas proved by Faenza, He, and Sethuraman for the bipartite matching polytope.

Hypothesis audit for the citation. Their Lemmas 21–22 apply to the matrix A of the bipartite matching polytope, i.e. exactly to the matrix $A = (I_n \mid A_G)$ considered here, with one loop column

per vertex and one valid-edge column per graph edge. Our B is a basis of this matrix. Hence the hypotheses match. (ar5iv.org)

By Lemma 21 of Faenza–He–Sethuraman, the valid edges E_B form a forest, and each connected component of the graph induced by B contains exactly one loop column. By Lemma 22, after permuting rows and columns, the basis matrix A_B becomes block upper triangular with diagonal entries all equal to 1. Therefore

$$\det(A_B) = \pm 1.$$

Since A_B is an integer matrix with determinant ± 1 , its inverse is again integral. Because $\mathbf{b} = \mathbf{1}$ is integral, we obtain

$$x_B^B = A_B^{-1} \mathbf{1} \in \mathbb{Z}^n.$$

As $x^B \geq 0$, every basic variable is a nonnegative integer. Moreover, every column of A has entries in $\{0, 1\}$, and every row equation has right-hand side 1, so each basic variable is at most 1. Hence

$$x_j^B \in \{0, 1\} \quad \text{for every } j \in B.$$

Let

$$M_B := \{e \in E_B : x_e^B = 1\}.$$

Because every row sum is exactly 1, no two edges of M_B can be incident to the same vertex; thus M_B is a matching in G . In particular,

$$|M_B| \leq \nu(G).$$

Now decompose the forest E_B into connected components T . Fix one such component T . Let

$$s_T := |M_B \cap E(T)|,$$

and let $\lambda_T \in \{0, 1\}$ be the value of the unique loop of that component in the basic solution x^B .

Every vertex of T is covered exactly once by the selected 1-valued columns of x^B : each 1-valued valid edge covers two vertices, and the loop (if $\lambda_T = 1$) covers one vertex. Therefore

$$|V(T)| = 2s_T + \lambda_T.$$

Since the valid edges of T form a tree,

$$|E(T)| = |V(T)| - 1 = 2s_T + \lambda_T - 1 \leq 2s_T.$$

Summing this inequality over all tree components gives

$$|E_B| = \sum_T |E(T)| \leq \sum_T 2s_T = 2|M_B| \leq 2\nu(G).$$

This is the desired bound. □

Lemma 2. *Let $r = \nu(G)$. The number of feasible bases of the bipartite matching polytope is at most*

$$\sum_{k=0}^{2r} \binom{m}{k} 2^k.$$

Proof. Fix a feasible basis B , and let $k := |B \cap E|$. By the previous lemma, $k \leq 2r$.

Conversely, to specify a feasible basis B , it is enough to specify:

1. the set $F := B \cap E$ of k valid edges, and
2. for every nontrivial connected component of the forest F , the vertex whose loop belongs to B .

Indeed, once the valid-edge forest F is fixed, every isolated vertex must contribute its own loop, and every nontrivial tree component must contribute exactly one loop by the forest-with-single-loops structure.

Now fix k . There are at most $\binom{m}{k}$ possibilities for F . Suppose F has nontrivial connected components $T_1, \dots, T_t$, and write $s_i := |E(T_i)|$. Then each T_i has $s_i + 1$ vertices, so the number of choices for the loop placement is

$$\prod_{i=1}^t (s_i + 1).$$

Since each $s_i \geq 1$, we have $s_i + 1 \leq 2^{s_i}$. Therefore

$$\prod_{i=1}^t (s_i + 1) \leq \prod_{i=1}^t 2^{s_i} = 2^{\sum_i s_i} = 2^k.$$

Hence the number of feasible bases with exactly k non-loop columns is at most $\binom{m}{k} 2^k$. Summing over $0 \leq k \leq 2r$ yields

$$\#\{\text{feasible bases}\} \leq \sum_{k=0}^{2r} \binom{m}{k} 2^k.$$

□

Lemma 3. *The number of possible Scarf pairs (B, D) for the input $(A, \mathbf{1}, C)$ is at most*

$$nN \sum_{k=0}^{2r} \binom{m}{k} 2^k.$$

Proof. A Scarf pair (B, D) has B a feasible basis, D an ordinal basis, and $|B \cap D| = n - 1$ unless $B = D$ and the algorithm has already terminated. For the nonterminal pairs relevant to counting iterations, D is determined by

$$D = B - \{j_\ell\} \cup \{j_t\},$$

where $j_\ell \in B$ and $j_t \notin B$.

Therefore, once B is fixed, there are at most n choices for j_ℓ and at most $N = n + m$ choices for j_t . Some such choices do not produce an ordinal basis, but this still gives a valid upper bound:

$$\#\{\text{Scarf pairs}\} \leq (\#\{\text{feasible bases}\}) \cdot nN.$$

Applying the previous lemma gives

$$\#\{\text{Scarf pairs}\} \leq nN \sum_{k=0}^{2r} \binom{m}{k} 2^k.$$

□

Proof of the theorem. First, at least one finite terminating execution exists. This is the classical perturbation-based degeneracy resolution for Scarf's algorithm: one perturbs the right-hand side to a nearby nondegenerate instance, runs the nondegenerate algorithm, and obtains a degeneracy-resolving cardinal-pivot rule for the original input. This is summarized explicitly in Faenza–He–Sethuraman, §3.3, as standard background from Scarf's original treatment.

Hypothesis audit for the citation. The perturbation principle requires precisely the standard hypotheses of Scarf's lemma: A in standard form with nonnegative entries, $b > 0$, C ordinal, and bounded feasible region. In our setting $A = (I_n \mid A_G) \geq 0$, $b = \mathbf{1} > 0$, C is ordinal by assumption, and $P(G)$ is bounded because every variable satisfies $0 \leq x_j \leq 1$. Hence the cited principle applies. (ar5iv.org)

Among all terminating executions, choose one with the minimum possible number of iterations. Let

$$(B_0, D_0), (B_1, D_1), \dots, (B_{T-1}, D_{T-1})$$

be the nonterminal Scarf pairs visited by this shortest terminating execution.

I claim that these Scarf pairs are pairwise distinct. Suppose instead that

$$(B_i, D_i) = (B_j, D_j) \quad \text{for some } 0 \leq i < j \leq T-1.$$

Because the state at times i and j is identical, the suffix of the execution starting immediately after time j can also be followed immediately after time i . Deleting the loop between times i and j therefore yields a strictly shorter terminating execution, contradicting minimality. Thus no Scarf pair repeats.

Consequently,

$$T \leq \#\{\text{possible Scarf pairs}\}.$$

By the previous lemma,

$$T \leq nN \sum_{k=0}^{2r} \binom{m}{k} 2^k.$$

Finally, since $m \leq N$,

$$\sum_{k=0}^{2r} \binom{m}{k} 2^k \leq \sum_{k=0}^{2r} (2m)^k \leq \sum_{k=0}^{2r} (2N)^k \leq (2r+1)(2N)^{2r}.$$

Therefore

$$T \leq nN(2r+1)(2N)^{2r} \leq (2r+1)2^{2r}N^{2r+2}.$$

This proves the existence of a terminating execution with the announced bound. If one counts elementary pivots rather than Scarf iterations, the bound simply doubles up to an additive 1. $\square$

Gap to the full statement

What remains open after the theorem above is substantial.

1. My result is *parameterized* by $r = \nu(G)$. When r grows with $n + m$, the bound

$$(2r+1)2^{2r}(n+m)^{2r+2}$$

is generally superpolynomial, so this does not resolve the unrestricted problem.

2. The theorem addresses the *existential execution* reading. It does *not* produce a single explicit, efficiently computable, instance-independent cardinal pivot rule that guarantees polynomial convergence for all inputs.
3. The proof uses the unit-capacity matching-polytope reading $b = \mathbf{1}$. I do not treat arbitrary-capacity b -matching/transportation variants.

So the full arbitrary- C problem on general bipartite matching polytopes is still unsettled by the present argument.

Self red-team

Counterexample attempts.

- I tried to break Lemma 1 with long alternating paths. On a rooted path with $2s$ or $2s + 1$ valid edges, the basic solution alternates $1, 0, 1, 0, \dots$, so $|E_B| = 2s$ or $2s - 1$, consistent with $|E_B| \leq 2|M_B|$.
- I tried branching trees. For the component with edges ra, aa', rb and loop at r , the feasible basic solution is $x_{aa'} = x_{rb} = 1$, $x_{ra} = x_{\ell_r} = 0$. Here $|E_B| = 3$, $|M_B| = 2$, again consistent.
- I checked the degenerate boundary case $\nu(G) = 0$: there are no valid edges, the only feasible basis is the identity basis, and the algorithm is already at a dominating basis if and only if the initial ordinal basis agrees with it. The theorem's upper bound is vacuous but correct.

Three likely referee objections and responses.

1. “*You used integrality of the basic solution too quickly.*” Response: I explicitly invoked the block-form lemma for basis matrices, deduced $\det(A_B) = \pm 1$, and then concluded A_B^{-1} is integral. Together with $b = \mathbf{1}$ and $0 \leq x_j \leq 1$, this yields $x_j \in \{0, 1\}$.
2. “*Your count of feasible bases may be missing exponential loop choices.*” Response: once the valid-edge forest is fixed, isolated vertices have forced loops, and a nontrivial component with s edges has only $s + 1$ root choices. The product bound $\prod (s_i + 1) \leq 2^{\sum s_i}$ controls this exactly.
3. “*Deleting repeated states only works if the pivot rule is memoryless.*” Response: the theorem is existential. I do not fix a global rule. If a terminating execution repeats a state, I may simply choose the later outgoing move when revisiting that same state and hence excise the loop. This is legitimate because I am only proving existence of *some* short execution.

Boundary checks.

- If the initial pair already satisfies $B = D$, then the iteration count is 0, consistent with the theorem.
- If a component has no valid edges, then its unique loop must be 1-valued; the component contributes 0 to both sides of $|E(T)| \leq 2s_T$.
- The proof never uses any preference-consistency property of C ; only the definition of Scarf pairs and the existence of some terminating execution are used.

Ideas tried

1. **Try to disprove the statement by importing exponential lower bounds from network simplex / transportation pivots.** Idea: because the bipartite matching matrix is closely related to network matrices, perhaps arbitrary C could force the same entering-column sequence as a bad network-simplex instance. Obstruction: the known lower-bound mechanisms I found do not directly control Scarf’s ordinally determined entering column, and I could not make the reduction robust against all admissible cardinal leaving choices.
2. **Try a conditional negative result via PPAD-hardness.** Idea: general SCARF is PPAD-complete, so perhaps arbitrary C on bipartite matching polytopes is already expressive enough to encode general SCARF instances. Obstruction: I found no restriction theorem in the literature nor an independent embedding showing PPAD-hardness persists under this specialization. Without such an embedding, the complexity-theoretic route stayed speculative.
3. **Try to extend the separator / disliked-edge machinery of Faenza–He–Sethuraman from their special matrix C^* to arbitrary C .** Idea: if a separator-like potential survived, one could hope for a direct polynomial monotonicity argument on general bipartite graphs. Obstruction: their arguments use the very special row structure of the consistent matrix C^* . For arbitrary C , the combinatorial meaning of “disliked edge” disappears, and I could not recover a monotone global potential.
4. **Count the reachable state space using only the matching-polytope combinatorics.** Idea: perhaps feasible bases on matching polytopes are much sparser than arbitrary bases. Outcome: this succeeded. The crucial observation is that a feasible basis contains at most $2\nu(G)$ non-loop columns. This yields a polynomial bound for fixed $\nu(G)$, and hence a rigorous partial result.

Invoked results

1. **Basis characterization for the bipartite matching polytope (T00L).** *Statement.* Let B be a set of n columns of the standard-form bipartite matching matrix A . Then B is a (possibly infeasible) basis if and only if the valid edges corresponding to B form a forest and each connected component contains exactly one loop. *Citation.* Faenza, He, Sethuraman (2025), Lemma 21, *Mathematics of Operations Research*, doi 10.1287/moor.2023.0055. (ar5iv.org) *Hypothesis-match audit.* Our $A = (I_n \mid A_G)$ is exactly the standard-form matrix of a bipartite matching polytope with one loop column per vertex; B is a set of n columns. Hence the lemma applies verbatim.
2. **Block form (hence unimodularity) of basis matrices (T00L).** *Statement.* For any basis B of the bipartite matching matrix A , the submatrix A_B can be permuted into a block upper-triangular form whose diagonal entries are all 1; in particular $\det(A_B) = \pm 1$. *Citation.* Faenza, He, Sethuraman (2025), Lemma 22, *Mathematics of Operations Research*, doi 10.1287/moor.2023.0055. (ar5iv.org) *Hypothesis-match audit.* Same as above: A is the standard-form matrix of a bipartite matching polytope, and B is a basis of A . Therefore the lemma applies.
3. **Perturbation-based resolution of degeneracy for Scarf’s algorithm (TOOL/BACKGROUND).** *Statement.* For a bounded instance (A, b, C) of Scarf’s algorithm

in standard form, there exists a way to resolve degeneracy by perturbing b to a nearby nondegenerate right-hand side and following the nondegenerate Scarf path; this yields at least one terminating execution to a dominating basis for the original degenerate instance. *Citation.* Classical background from Scarf–Hansen (1973), summarized in Faenza, He, Sethuraman (2025), §3.3. (cowles.yale.edu) *Hypothesis-match audit.* Our instance satisfies the standard hypotheses: A is in standard form and nonnegative, $b = \mathbf{1} > 0$, C is ordinal, and $P(G)$ is bounded because $0 \leq x_j \leq 1$ for every variable. Hence the perturbation principle applies.

Self-confidence

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I checked the counting argument carefully and could not break it; my only residual caution is that I relied on the classical perturbation-based existence of some terminating execution rather than reproving that background fact from first principles.

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